Highlights

Right Temporal EEG Biomarkers for tDCS Remission: A Parsimonious Deep Learning Approach to Precision Psychiatry

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- Right temporal (TP10) electrode activity emerges as the dominant biomarker for tDCS remission prediction, with 4 of 5 optimal features originating from this region.
- Feature parsimony triumphs: a minimal set of 5 features decisively outperforms models using all 246 features, demonstrating that “less is more” in biomarker discovery.
- Hybrid CNN-LSTM achieves perfect precision (100%) with clinically valuable recall (57.14%), supporting precision-oriented treatment selection using portable 4-channel EEG.

Right Temporal EEG Biomarkers for tDCS Remission: A Parsimonious Deep Learning Approach to Precision Psychiatry

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ABSTRACT

Identifying reliable biomarkers for predicting treatment response in major depressive disorder (MDD) remains a critical challenge in precision psychiatry. While transcranial direct current stimulation (tDCS) shows promise as a non-pharmacological intervention, clinical response is highly variable. This study reveals a striking neuroanatomical signature: right temporal (TP10) electrode activity dominates predictive power for tDCS remission. Using four-channel resting-state EEG from 21 MDD patients, we demonstrate that a minimal set of five features—four from the right temporal region and one capturing frontal-temporal beta asymmetry—outperforms models using all 246 extracted features. A hybrid one-dimensional Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM) architecture, evaluated using leave-one-patient-out cross-validation, achieved 85.71% accuracy with perfect precision (100%) and 57.14% recall at the patient level. This "less is more" finding—where feature parsimony decisively beats feature abundance—identifies right temporal activity as a key biomarker for tDCS response. The results demonstrate that portable, low-density EEG combined with targeted feature selection can create a reliable, non-invasive tool for personalized depression treatment, revealing that the signal lies not in complexity but in the right anatomical location.

Introduction

Major depressive disorder (MDD) affects over 280 million people worldwide, representing one of the leading causes of disability and contributing significantly to the global burden of disease [33]. According to the World Health Organization, depression is projected to become the top contributor to disease burden by 2030, emphasizing the urgent need for effective treatment approaches [23]. Major Depressive Disorder (MDD) is characterised by prolonged low mood, diminished interest, exhaustion, disrupted sleep patterns, and feelings of unease [4] However, its heterogeneity, meaning it exhibits a wide range of symptom clusters, and its aetiology is not attributed to any specific assumptions. When diagnosing and treating MDD, it is essential to take into account all aspects of the condition, understanding that improvements in certain symptoms such as mood may occur before improvements in other symptoms like sleep disturbance [20]. MDD has a substantial impact on the quality of life, specifically in terms of health-related aspects. The impact of treatment response on the quality of life is a growing concern in patient care [6]. The current pharmacological treatments have limited effectiveness, resulting in a high occurrence of residual symptoms and side effects. This emphasises the necessity for the creation of innovative therapies and approaches [29].

EEG data is analysed using a range of mathematical tools and techniques, such as spectral analysis and wavelet transform. These techniques aid in identifying distinct patterns in EEG data [28]. Technological improvements, such as the development of wearable EEG devices, have improved the practicality and usefulness of technology throughout time [8]. The capacity to record alterations in cerebral function and composition, along with its straightforwardness and lack of invasiveness, renders it a desirable instrument in diverse medical and scientific contexts [25].

Transcranial direct current stimulation (tDCS) is emerging as a promising non-invasive neurostimulation technique, offering significant therapeutic benefits with minimal side effects compared to traditional pharmacological interventions [5]. tDCS delivers low-amplitude direct current (typically 1-2 mA) to the scalp via electrodes, modulating cortical excitability in targeted brain regions. These currents have the ability to inducelead to neuromodulatory changes in cortical activity that persist beyond the duration of the stimulation [5] In depression treatment, tDCSthe anode is commonly applied placed over the left dorsolateral prefrontal cortex with anodal stimulation on the left hemisphere and cathode on al stimulation on the right to rebalance neural activity associated with mood regulation [29]. Recent meta-analyses demonstrate tDCS's clinical efficacy in depression treatment [30]. A meta-analysis of 23 randomized controlled trials in depressive episoded, involving 1,092 participants, found that active tDCS was superior to sham stimulation, with a moderate effect size (Hedges' g $\approx$ 0.46) in depression severity and

Abbreviations: MDD, Major Depressive Disorder; tDCS, Transcranial Direct Current Stimulation; EEG, Electroencephalogram; CNN, Convolutional Neural Network; LSTM, Long Short-Term Memory; PSD, Power Spectral Density; AUC, Area Under the Curve; LOPO-CV, Leave-One-Patient-Out Cross-Validation.

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approximately doubled rates of clinical response (33% vs 16%) and remission (19% vs 10%), while discontinuation rates showed no significant difference between active and sham treatment arms [35]. Individual participant interviews have also shown high acceptability for tDCS treatment [37, 36]. These findings show that tDCS is an effective and safe potential treatment with high acceptability, evidenced by similar discontinuation rates to sham conditions and qualitative assessments A fully remote, multisite, double-blind, placebo-controlled randomized trial of home-based tDCS for MDD (n=174). active stimulation was associated with significantly greater improvements in depressive symptoms, as measured by the Hamilton Depression Rating Scale than sham (-9.41 ± 6.25 vs -7.14 ± 6.10; 10-week means 9.58 ± 6.02 vs 11.66 ± 5.96; 95% CI = 0.51–4.01; $P = 0.012$), with no differences in discontinuation rates.

Woodham et al. [42] clinical trial observed no serious adverse events related to tDCS device usage, and no participants developing mania or hypomania. However higher rates of skin redness were observed in the active stimulation treatment group as compared to the inactive stimulation group, while there were no significant differences in other adverse events. While tDCS is generally safe, careful attention to proper device use and monitoring of adverse events is necessary for ensuring safety. These findings support home-based, remotely supervised tDCS as an efficacious, acceptable, and safe antidepressant intervention [42].

An electroencephalogram (EEG) quantifies the cerebral electrical activity by means of electrodes positioned on the scalp EEG is extensively utilised in both clinical and scientific contexts due to its broad variety of applications. The EEG signals serve as reliable indicators of the brain's health conditions, offering vital insights for the diagnosis of neurological disorders as well as for monitoring the level of anaesthesia during surgical procedures [40].

Complementing clinical findings, baseline four-channel EEG connectivity (phase-locking value; PLV) from the same home-tDCS clincal trial Woodham et al. [42] showed (i) higher gamma PLV with active vs. sham and (ii) that combined theta/alpha/beta PLV features could predict remission (~72% accuracy) using deep learning [46].

The development of reliable biomarkers for predicting tDCS treatment response represents a critical step toward precision psychiatry, potentially transforming clinical decision-making and improving patient outcomes. Early identification of responders could help prioritize tDCS for appropriate patients while directing non-responders sooner to alternative therapies, thereby embodying a personalized medicine approach in depression treatment.

The application of deep learning architectures to EEG analysis has demonstrated superior performance compared to traditional machine learning approaches, with hybrid Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM) models achieving 88-99% accuracy in various EEG classification tasks [1, 38]. These architectures combine the spatial feature extraction capabilities of CNNs with the temporal modeling strengths of LSTMs, making them particularly well-suited for analyzing the complex spatiotemporal patterns inherent in EEG signals.

A distinguishing characteristic of EEG signals is their rich temporal structure, reflecting dynamic brain oscillations and events. Traditional EEG biomarkers for depression response have often been computed as static features such as average band power or coherence over a recording epoch which may overlook how brain activity evolves over time [17]. Temporal sequence analysis techniques, such as recurrent neural networks, can potentially detect complex temporal patterns or transient events that distinguish responders from non-responders. Recent advances in deep learning have made it possible to apply temporal sequence models to EEG data to automatically learn discriminative patterns across time that human observers or simpler algorithms might miss [32].

Existing EEG studies employ high-density electrode systems with 19-128 channels, creating significant barriers to clinical deployment due to complexity, cost, and patient compliance issues [41]. The development of effective prediction models using reduced electrode configurations represents a critical need for translating research findings into clinical practice.

We present the first investigation applying a hybrid one-dimensional CNN-LSTM architecture specifically to predict tDCS treatment remission in MDD patients. This represents a significant methodological advancement over existing traditional machine learning approaches in the sparse tDCS prediction literature.

Beyond single-modality markers, converging evidence suggests that MDD comprises distinct *neuroanatomical dimensions* with different treatment propensities, identified at scale with structural MRI using semi-supervised clustering (HYDRA) [13]. In particular, one dimension with largely preserved volumes showed greater improvement with SSRIs than placebo, whereas a second dimension with subtle, widespread reductions improved similarly with either SSRIs or placebo [13]. These results motivate a *multimodal* strategy in which compact EEG biomarkers (e.g., PSD and asymmetry) are linked to macro-scale MRI signatures to achieve dimension-aware prediction of treatment response.

This paper details our approach and findings. Section 1 reviews the relevant literature on tDCS response prediction and EEG analysis. Section 2 outlines the methodology, including participant data, preprocessing, feature extraction, and the model architecture. Section 3 presents the results of our model, including performance metrics and feature importance. Section 4 provides a discussion of the results, their implications, and the limitations of the study. Finally, Section 5 concludes the paper with a summary of our findings and their potential impact on personalized depression treatment.

1. Related Work

1.1. Predicting Response to tDCS in Major Depressive Disorder

Al-Kaysi et al. [3] noted the practicality of employing machine learning to identify patients who are likely to benefit from tDCS treatment, by considering factors such as sample size and diversity, though the sample size was limited. Development and validation of machine learning algorithms, such as support vector machine (SVM), linear discriminant analysis (LDA), and extreme learning machine (ELM), can effectively classify EEG signals for tDCS response prediction. Enhancements in the precision and resilience of algorithms are imperative for the practical implementation in clinical settings [2]. Dagnino et al. [10] emphasised the need for replication and validation of the current findings in larger and more diverse groups of patients. Ensuring the broad applicability and transferability of prediction models across diverse clinical settings is crucial. To enhance the reliability and validity of future research in this area, it is crucial to address limitations such as the absence of sham conditions in certain studies, limited electrode arrays, and small sample sizes [47]. Integration with Clinical Data: It is necessary to examine how data of different modalities and domain can be efficiently combined using machine learning methods to achieve a comprehensive prediction of tDCS treatment response [27]. Prioritising the identification of distinct biomarkers present in EEG, clinical, and neuropsychological data that can consistently and accurately forecast the response to treatment. This entails investigating various EEG characteristics and comprehending their correlation with clinical outcomes [19]. Complementary to EEG studies, large multisite structural MRI analyses have delineated two robust *dimensions* in MDD with distinct treatment implications [13]. Using HYDRA on harmonized raw MRI from medication-free patients, Fu et al. [13] identified a dimension with preserved gray/white matter volumes (showing greater improvement with SSRIs than placebo) and a second dimension with subtle, widespread volumetric reductions (showing comparable improvement to SSRIs and placebo). This dimensional framing provides a principled context for integrating EEG with MRI to stratify patients and to test whether EEG predictors express differently across neuroanatomical dimensions.

Across home-tDCS cohorts in unipolar MDD and bipolar depression, low-density PLV connectivity features with deep learning have achieved ∼69–72% accuracy, underscoring the value of connectivity beyond band-power features [46, 45].

1.2. EEG Biomarkers for Treatment Response in Depression

Not all patients respond equally to tDCS, and EEG-based biomarkers offer a non-invasive, low-cost, and accessible means to potentially stratify participants who will likely show a clinical response from those with a lower likelihood before treatment begins which would prevent the initiation of a treatment that would not be beneficial and could delay initiation of a more appropriate treatment (ref). Investigation of the EEG characteristics (such as power spectral density, coherence, fractal dimensions) that have the highest predictive value for tDCS treatment response would aid in personalising treatment [3, 14, 15]. This entails examining various frequency bands and EEG channels to determine their predictive efficacy [19]. Puthankattil and Joseph [34] and Khadidos et al. [21] emphasize the importance of certain frequency-domain characteristics, such as Relative Wavelet Energy (RWE) and band power analysis, in differentiating EEG signals between normal and depressive states. There is potential for investigating further frequency-domain characteristics and their significance in relation to EEG patterns associated with depression and examining how different feature sets affect classification performance. Ghiasi et al. [16] and Zhao et al. [48] emphasize the significance of spatial information and microstate analysis in understanding the characteristics of EEG signals in individuals with depression. They propose the possibility of further investigation into the spatial dynamics of EEG features and their contribution to the identification of subclinical depression. In unipolar MDD, PLV-based deep models trained on pre-treatment four-channel EEG predicted tDCS remission with 71.9% accuracy (sensitivity 52.9%, specificity 83.1%), underscoring the usefulness of connectivity features in low-density, home-acquired EEG [46]. Additionally, Mahato et al. [22] emphasize the importance of artifact correction and precise preprocessing techniques in improving the quality and dependability of EEG data for detecting depression. Subsequent investigations could prioritize the enhancement of artifact removal techniques and examine their influence on the accuracy and resilience of feature extraction and classification. Additional investigation is required to explore the nonlinear features of EEG and their potential in predicting the outcomes of tDCS treatment. These features can capture more intricate aspects of brain activity linked to the response to treatment [12]. Longitudinal studies can be conducted to track changes in nonlinear EEG features before, during, and after tDCS treatment sessions. This approach could provide insights into how specific nonlinear EEG patterns evolve in response to tDCS and their correlation with treatment outcomes. The role of different tDCS montage configurations on nonlinear EEG features and treatment response can be explored. Bespoke machine learning algorithms can be used to identify complex patterns in nonlinear EEG data associated with tDCS treatment response, facilitating the development of predictive models for personalized treatment planning. Multimodal neuroimaging approaches, such as functional magnetic resonance imaging (fMRI) or positron emission tomography (PET), can be combined to provide a more comprehensive view of brain activity changes induced by tDCS and their relationship to nonlinear EEG features.

1.3. Machine Learning and Temporal Sequence Analysis for EEG

With the advent of machine learning particularly deep learning researchers increasingly complement curated EEG features with automated representation learning from raw or minimally processed signals. Classical approaches (e.g., support vector machines, linear discriminant analysis, logistic regression) typically operate on extracted features such as band power, coherence, or ERP amplitudes, and have been used to classify treatment responders versus non-responders.

Wu et al. [44] proposed Sparse EEG Latent Space Regression (SELSER), using a broad spectral feature set to predict antidepressant outcomes as a *regression* task, reporting cross-validated R^2 up to 0.60 on a large trial dataset (309 patients) with independent-sample testing.

Zhdanov et al. [49] trained an SVM on baseline EEG from 122 escitalopram patients using multiple feature families (spectral power, entropy, microstates), reporting *classification* accuracy of ∼79% for responder prediction; incorporating early-treatment EEG (week 2) increased accuracy to ∼82%. While this suggests temporal evolution may add information, improvements can also reflect protocol and cohort effects.

Hybrid deep models that combine convolutional encoders with recurrent sequence modeling have also been explored. Adamson et al. [1] used a 1D CNN followed by an LSTM for rTMS response classification and reported very high accuracies (up to 99% in sex-specific models) under their internal evaluation; such results should be interpreted in light of cohort size, split strategy (e.g., patient-wise vs. epoch-wise), and external validation. Similarly, Shahabi et al. [38] used wavelet time–frequency representations with a convolutional recurrent network and attention, reporting ∼80–90% accuracy across a private cohort (46 patients) and a larger public dataset (124 patients), with LSTM-based sequence modeling improving over static CNN features under their protocol.

In our study (Fig. 2), the convolutional pathway is followed by global average pooling over time; the subsequent BiLSTM and attention therefore operate on a sequence of length 1 and act primarily as gated/channel-wise mixers rather than long-range temporal models. We include them for their regularization and representational benefits, and we evaluate at the *patient level* using LOPO cross-validation to facilitate fair comparison across patients. Because prior work varies in endpoints (medication vs. rTMS), preprocessing, and evaluation protocols (metrics, splits, external validation), numerical results are not directly comparable across studies.

2. Methodology

2.1. Participants

This study utilized EEG data from participants diagnosed with major depressive disorder who participated in a fully remote, multisite, double-blind, placebo-controlled, randomized superiority trial investigating home-based transcranial direct current stimulation (tDCS) for treating MDD [42]. The study was conducted by the University of East London School of Psychology in collaboration with clinical partners. All participants provided written informed consent. Ethical approval was granted by the South Central–Hampshire B Research Ethics Committee (UK). Comprehensive clinical and demographic details for the trial are reported in Woodham et al. [42]. Participant characteristics are summarized in Table 1.

Participants were adults aged 18 years or older, diagnosed with major depressive disorder and in a currenta depressive episode of moderate severity or higher, as indicated by a score of 16 or more on the 17-item Hamilton Depression Rating Scale (HDRS), following DSM-5 criteria. Recruitment encompassed regions in England, Wales, and Texas, USA. Inclusion criteria required participants not to be resistant to depression treatments or on medications affecting cortical excitability. Exclusion criteria included inadequate response to antidepressant treatments, significant suicide risk, comorbid psychiatric disorders, or use of medications influencing cortical excitability. Written informed consent and access to video conference for remote session supervision were mandatory requirements. The study demographic comprised 21 participants who completed the full protocol and had usable EEG data. Following the 10-week tDCS intervention period, participants were classified into remission (n=7, 33.3%) and non-remission (n=14, 66.7%) groups based on standardized clinical assessment criteria. This distribution reflects typical clinical response rates observed in tDCS trials for depression treatment.

2.2. EEG Data Acquisition and Preprocessing

EEG data were collected during resting state conditions with participants' eyes closed, using a standardized four-electrode configuration (AF7, AF8, TP9, TP10) at a sampling rate of 256 Hz. The EEG signals were recorded in 10-second epochs, with some participants having recordings of 10 minutes duration while others had data split into two 5-minute files. The preprocessing pipeline used custom Python scripts and standard signal processing libraries. To standardize the data and correct slight window-size inconsistencies, all segments were resampled to a uniform length. Signals were then high-pass filtered at 0.5 Hz to remove slow drifts and low-pass filtered at 60 Hz to suppress high-frequency artifacts. Finally, each channel was upsampled, passed through a low-pass Butterworth filter, and downsampled to a predetermined length so that all segments were aligned to the same size. EEG window rejection used the device-provided Horse Shoe Indicator (HSI), which scores electrode contact quality per sample (1=good, 2=average, 4=poor). For each 10 s window, we averaged HSI values over samples (per electrode) and retained windows only when the average HSI was <2 (i.e.,

Table 1
Demographic and Clinical Characteristics of Participants

Characteristic	Value
Sample size (n)	21
Gender (Female/Male)	18/3
Age (years, mean ± SD)	37.1 ± 9.7
Inclusion criteria	Age ≥18; DSM-5 MDD; HAMD ≥16; MINI v7.0.2 confirmed
Medication/psychotherapy	Med-free ≥6 weeks or stable antidepressant; psychotherapy stable ≥6 weeks
Exclusion criteria	Mania/psychosis; treatment-resistant depression; neurological/medical confounds; prior admission for depression/suicidality; standard tDCS exclusions
Recruitment	Online advertisements; general practitioner referrals
Ethics	Written consent; South Central–Hampshire B REC (UK)

Comprehensive clinical and demographic details are reported in Woodham et al. [42].

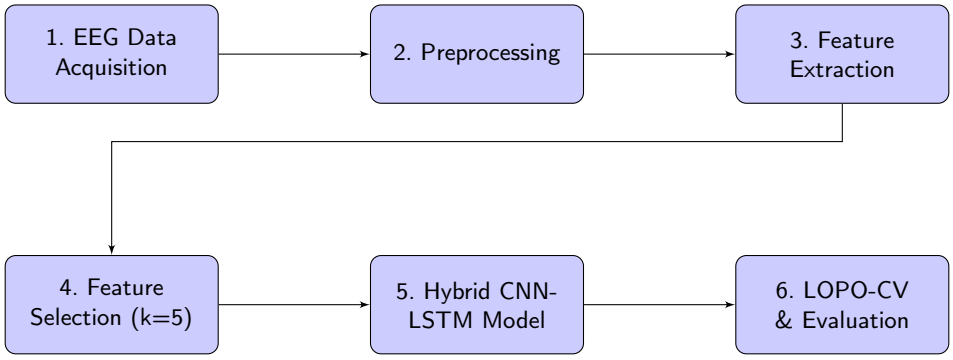


Figure 1: The methodology flowchart, from data acquisition to performance evaluation.

better than "average" contact); all other windows were discarded to avoid poor-contact artefacts.

2.3. Feature Extraction

We extracted features per channel across several signal domains using a configurable pipeline that emphasizes reproducible, lightweight measures aligned with our four-electrode montage. Features span spectral, temporal, entropy/complexity, asymmetry, cross-hemispheric, and simple inter-channel correlation summaries. This approach focuses on measures directly computed from Welch power spectral density (PSD) and basic time-series statistics, avoiding advanced graph-theoretic or phase-based connectivity metrics.

2.3.1. Spectral Features

Spectral features were derived from per-channel PSDs computed via Welch's method. For each canonical band (delta, theta, alpha, beta, gamma; as defined in the configuration), we computed absolute band power and total power, then derived relative band powers by normalizing each band by total power. We further computed frequency-weighted power, spectral entropy from the PSD, spectral edge frequencies (SEF-90 and SEF-95), and PSD summary statistics (mean, standard deviation, maximum, minimum, median, skewness, and kurtosis).

2.3.2. Temporal Features

Time-domain features included basic statistics per channel (mean, standard deviation, variance, skewness, kurtosis, root-mean-square, peak-to-peak amplitude, zero-crossings, and mean absolute deviation) alongside Hjorth parameters (activity, mobility, and complexity) computed from the signal and its derivatives.

2.3.3. Nonlinear and Entropy Features

Nonlinear and entropy-oriented measures were computed per channel to capture complexity beyond second-order statistics. We calculated sample entropy and spectral entropy, as well as correlation dimension, Hurst exponent, largest Lyapunov exponent, and detrended fluctuation analysis (DFA).

2.3.4. Inter-channel Correlation Features

Between-channel summaries were limited to simple, robust measures. For predefined channel pairs (AF7–AF8, TP9–TP10, AF7–TP9, AF8–TP10, AF7–TP10, AF8–TP9), we computed the maximum absolute cross-correlation (on z-scored signals) and the Pearson correlation coefficient as proxies for coupling.

2.3.5. Asymmetry Features

Given the importance of hemispheric balance in depression, we computed per-band log-power asymmetry for frontal (AF7–AF8) and temporal (TP9–TP10) pairs, along with ratio-based asymmetries. Cross-hemispheric

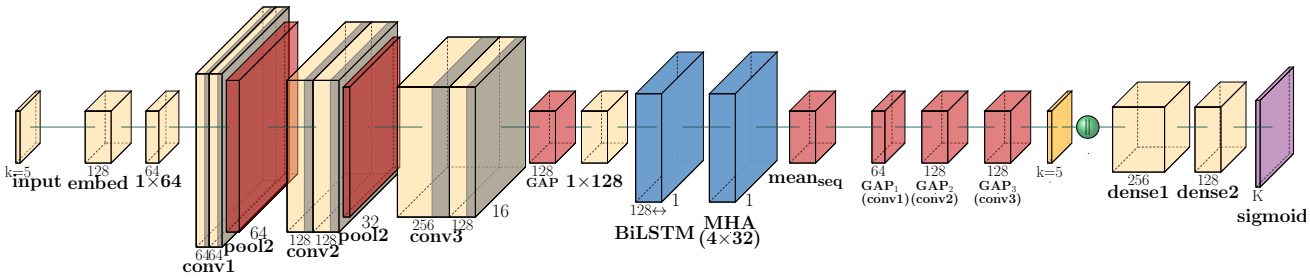


Figure 2: CNN–BiLSTM–Attention architecture

features included hemispheric asymmetry using averaged frontal–temporal powers (left: AF7/TP9 vs. right: AF8/TP10), frontal–temporal ratios and differences on each side, and diagonal frontal–temporal ratios/differences (AF7–TP10 and AF8–TP9).

2.3.6. Feature Selection and Dimensionality Reduction

In our experiments, feature selection for model training and evaluation used a univariate filter approach: **SelectKBest** with the **ANOVA F-statistic (*f_classif*)** as the scoring function. We evaluated two values of k $k = 5$ (primary) and $k = 20$ (exploratory) and also a *no selection* condition (all 246 features). The $k = 5$ configuration consistently yielded the best downstream patient-level performance and most clinically desirable error profile across models (see Section 3).

2.4. Hybrid CNN–LSTM with Lightweight Attention

Figure 2 shows the end-to-end pipeline. Tabular EEG features are embedded from n inputs to 128, projected to 64, and reshaped to a pseudo-sequence of length 64 for convolutional processing; this keeps the model efficient while enabling hierarchical feature learning.

Three sequential 1D convolutional blocks with ReLU and intermittent max-pooling reduce the temporal extent $64 \rightarrow 32 \rightarrow 16$: Block 1 uses 64 filters (kernels 5 and 3, pool 2), Block 2 uses 128 filters (kernels 3 and 3, pool 2), and Block 3 uses 256 then 128 filters (kernels 3 and 1, no pool). Batch normalization and progressive spatial dropout are applied throughout.

After GAP_{CNN} the 128-D vector is unsqueezed to 1×128 and passed to a two-layer bidirectional LSTM (128 units per direction with sequence output, then 64 per direction with final-state output) followed by multi-head attention (4 heads, key dimension 32); because the sequence length is 1, these layers act primarily as gated/channel-wise mixers rather than long-range temporal models, but they consistently improved representation quality and optimization.

We concatenate the post-attention vector with multi-scale taps (GAP_1: 64, GAP_2: 128, GAP_3: 128) and optional handcrafted features d (concat "‖"), then apply two dense layers (256 and 128 units with ReLU, BN, dropout) and a final sigmoid for K-way classification.

As shown in Fig. 2, the preselected handcrafted feature vector (k=5) is also injected via a skip pathway that bypasses the convolutional and recurrent blocks and is concatenated ("‖") with the deep features before the dense head. This preserves salient, interpretable signals and acts like a residual shortcut in feature space, complementing the learned representation.

From left to right: preselected handcrafted features from preprocessing (k=5, SelectKBest) → linear embedding to 128 ("embed") → projection to 64 and unsqueeze to a pseudo-sequence of length 64 ("1×64") → three 1D Conv blocks with temporal extents $64 \rightarrow 32 \rightarrow 16$ (Block 1: $[64, 64]$, k=5, 3, pool 2; Block 2: $[128, 128]$, k=3, 3, pool 2; Block 3: $[256, 128]$, k=3, 1, no pool) → global average pooling over time to 128 (GAP_{CNN}) and unsqueeze to 1×128 → BiLSTM (128↔, 64↔) and multi-head attention (4×32) operating on a single step, followed by mean over the sequence (len= 1) → feature-pyramid vectors $\text{GAP}_1/\text{GAP}_2/\text{GAP}_3$ with sizes 64/128/128 plus a handcrafted skip (k=5) → concatenation (‖) → dense layers 256 and 128 (ReLU+BN+dropout) → K-way sigmoid output. *Notes:* time labels denote effective pseudo-time after pooling; because the recurrent/attention block sees seq. len= 1, it acts primarily as gated/channel-wise mixing. Dropout: CNN blocks 0.10/0.20/0.30; BiLSTM 0.20/0.10 (layer 1 input/recurrent) and 0.30/0.20 (layer 2); attention 0.10; dense 0.30/0.40.

2.5. Model Training and Evaluation

Models were trained with batch size 1024 using AdamW (initial learning rate 0.001, weight decay 0.01, β_1=0.9, β_2=0.999, ϵ=10^{-7}) and a cosine-annealing schedule with warm restarts (first cycle length 20, multiplier 1, minimum learning rate 10^{-6}). Class imbalance (14 non-remission, 7 remission) was addressed via inverse-frequency class weights. Regularization comprised L2 weight decay, label smoothing (0.05), spatial dropout in convolutional layers, and recurrent dropout in LSTMs; early stopping monitored validation macro F1 (patience 10) with best-weight restoration. We used mixed precision on an NVIDIA RTX 3090 and fixed random seeds for reproducibility.

We used leave-one-patient-out (LOPO) cross-validation. In each fold, one patient was held out exclusively for testing while the model was trained on the remaining patients; an internal validation set drawn only from the training patients guided early stopping, preventing any information leakage. For the held-out patient, segment-level probabilities were averaged to obtain a patient-level decision per fold, and performance was summarized across folds.

We report macro-averaged accuracy, precision, recall, F1-score, and area under the ROC curve (AUC–ROC) together with patient-level confusion matrices, aggregated over LOPO folds.

Because the time axis is globally pooled before the recurrent/attention block, the BiLSTM and multi-head attention operate on a single-step sequence and function primarily as gated/channel-wise mixers rather than long-range temporal models; we include them for their demonstrated gains in representation quality and optimization, and leave full temporal modeling (with longer pre-LSTM sequences) to future work. Feature selection (SelectKBest with $f_classif$) was performed once during preprocessing on the full labeled dataset prior to cross-validation; this can introduce optimistic bias because labels from held-out patients may influence feature ranking. Future work will perform fold-specific selection or use a separate development set to avoid leakage.

3. Results

3.1. Model Performance

The hybrid 1D CNN-LSTM model achieved robust performance for predicting tDCS treatment remission using the four-electrode EEG configuration and five selected features. Performance was evaluated using Leave-One-Patient-Out cross-validation to provide conservative estimates suitable for clinical translation assessment.

3.1.1. Patient-Level Classification Results

At the patient level, the model demonstrated strong overall performance with 85.71% accuracy (18 out of 21 patients correctly classified). The confusion matrix analysis revealed excellent specificity with zero false positive predictions, resulting in perfect precision (100%) for remission prediction. However, sensitivity was more moderate at 57.14%, with 4 true positive predictions out of 7 actual remission patients.

The model achieved an F1-score of 72.73%, reflecting the balance between perfect precision and moderate recall. The area under the ROC curve (AUC) was 76.53%, indicating good discriminative ability between remission and non-remission groups. These results represent competitive performance given the small sample size and conservative validation approach (Figure 3; Table 2).

Table 2
Performance of the Hybrid 1D CNN-LSTM Model (5 selected features)

Metric	Value (%)
Accuracy	85.71
Precision	100.00
Recall	57.14
F1-Score	72.73
ROC-AUC	76.53
Confusion Matrix	
True Negatives	14
False Positives	0
False Negatives	3
True Positives	4

3.1.2. Performance Characteristics

The model's performance profile indicates conservative prediction behavior, favoring specificity over sensitivity. This characteristic may be clinically advantageous, as false positive predictions (incorrectly predicting remission) could lead to inappropriate treatment decisions, while false negatives may result in patients receiving tDCS who might benefit from alternative treatments.

The perfect precision score demonstrates that when the model predicts remission, this prediction is consistently accurate. The moderate recall indicates that the model successfully identifies 57% of patients who will achieve remission, while conservatively classifying the remaining remission patients as non-remission to avoid false positive errors.

3.2. Feature Selection and Importance

Feature importance analysis revealed the relative contribution of each selected EEG feature to the classification decision. The analysis utilized both permutation importance and attention weight visualization to identify the most predictive neural signatures for tDCS remission (Figure 5).

3.2.1. Selected Features

Feature importance analysis, using the ANOVA F-statistic, identified a compact and powerful set of predictors. While a broader set of 20 features spanning various spectral, coherence, and entropy domains was evaluated, a focused set of the top five features yielded the strongest model performance. This optimal set was dominated by metrics from the right temporal electrode (TP10), including its total power, and the mean, standard deviation, and maximum of its power spectral density. The fifth feature was the beta-band difference between left-frontal and temporal regions, highlighting the importance of both localized temporal activity and inter-regional asymmetry in predicting tDCS remission.

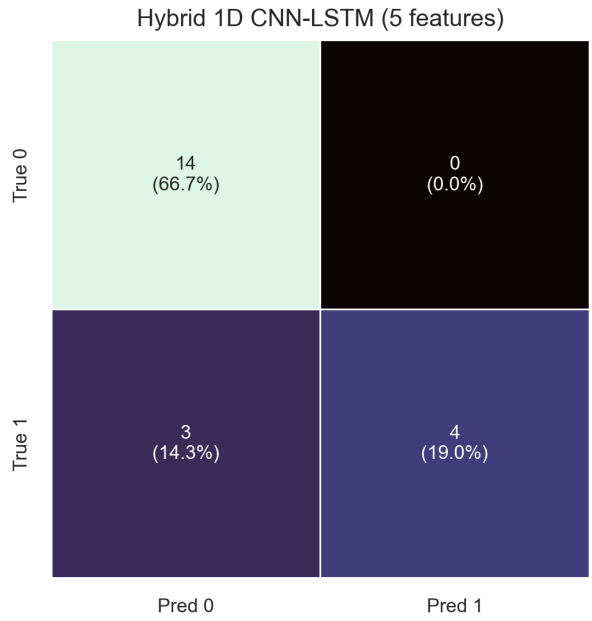


Figure 3: Hybrid 1D CNN-LSTM confusion matrix (5 selected features).

3.2.2. Feature Selection Validation via Downstream Performance

We validated the five-feature SelectKBest (*f_classif*) selection through *downstream patient-level performance* across multiple architectures rather than via post hoc importance scores. With the same five selected features, several models achieved their best results (Table 3): notably, SVM (Linear) reached **F1 75.00%** and **ROC-AUC 89.80%**, while the hybrid CNN-LSTM achieved **F1 72.73%** with **100% precision**. Expanding to 20 features degraded performance markedly (Table 5; e.g., SVM Linear F1 15.38%, AUC 36.73%; hybrid F1 20.00%), and using all 246 features degraded further (Table 7; SVM Linear F1 16.67%, AUC 34.69%; hybrid F1 18.18%). This consistent, cross-model superiority with the five selected features provides empirical validation that SelectKBest(*f_classif*) at $k = 5$ is optimal for this dataset and clinical objective.

3.2.3. Neurophysiological Interpretation

The dominance of TP10-related features suggests that right temporal region activity patterns are particularly informative for predicting tDCS response. This finding aligns with previous research indicating the importance of temporal lobe activity in depression and treatment response mechanisms.

The significant contribution of left frontal-temporal beta band asymmetry (left_frontal_temporal_diff_beta) highlights the role of inter-regional connectivity patterns in treatment prediction. Beta band activity has been associated with cognitive control and attention regulation, processes that may be critical for tDCS treatment mechanisms.

3.3. Comparative Performance Across Models (5 selected features)

To contextualize the hybrid CNN-LSTM performance, we compared results with alternative architectures trained using the same five-feature configuration and patient-level evaluation. Summary metrics are provided in Table 3 with bar-chart visualization in Figure 4a. Patient-level confusion matrices are listed in Table 4.

3.3.1. Interpretation

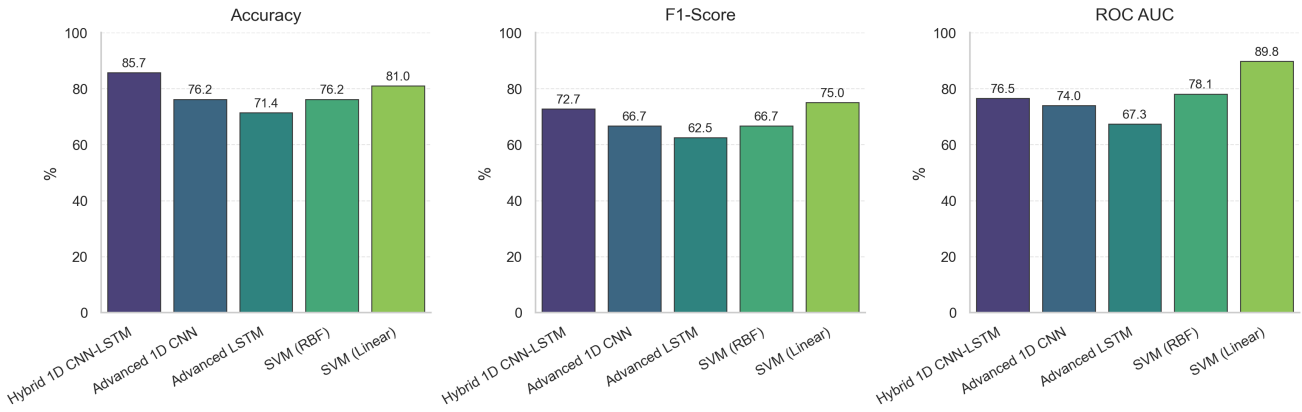
The SVM with a linear kernel achieved the highest F1-score (75.00%) and ROC-AUC (89.80%), while the hybrid CNN-LSTM achieved the highest accuracy (85.71%) and perfect precision (100%) due to zero false positives. This conservative prediction profile is clinically advantageous for avoiding inappropriate treatment allocation.

3.4. Comparative Performance Across Models (20 selected features)

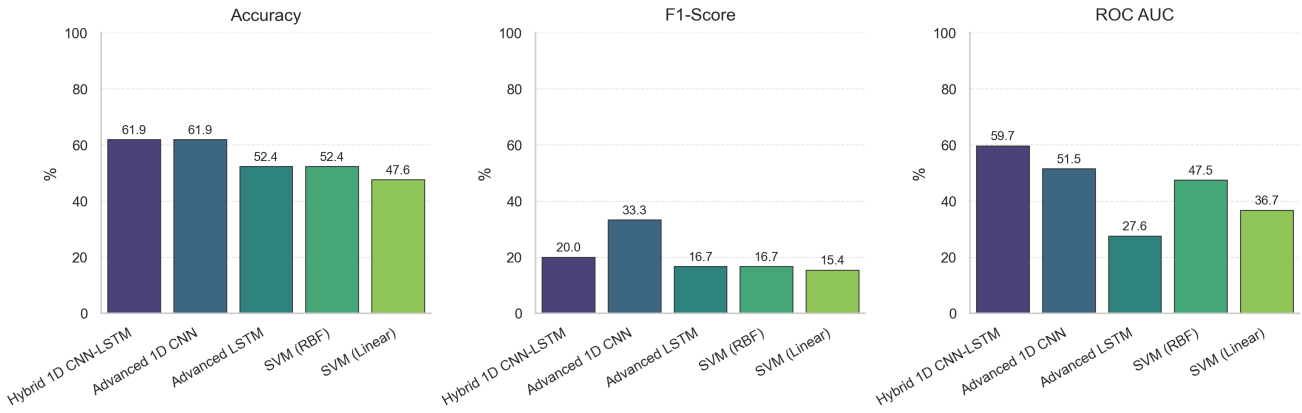
We also evaluated all models with 20 selected features. Performance declined relative to the 5-feature condition but models generally remained functional. Summary metrics are shown in Table 5 with a visual overview in Figure 4b. Confusion matrices are provided in Table 6.

3.5. Impact of Feature Selection (5 vs. 20 vs. 246 features)

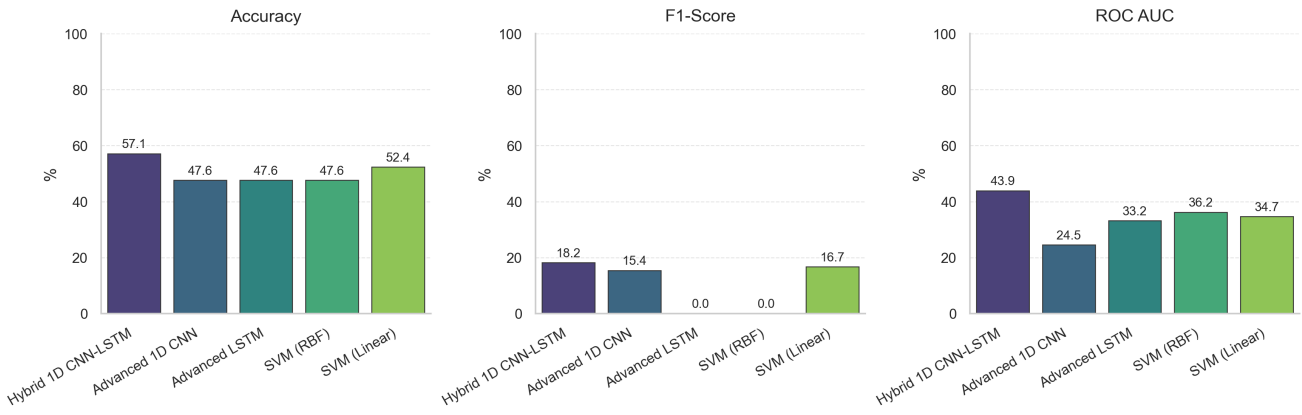
We trained the same set of models without feature selection using all 246 features. Performance degraded substantially across all models, with two models failing to identify any responders. The 20-feature setting performs better than 246 features but remains clearly inferior to the



(a) Model comparison with 5 selected features.



(b) Model comparison with 20 selected features.



(c) Model comparison without feature selection (246 features).

Figure 4: Model comparison with 5, 20, and 246 features.

Table 3
Patient-level performance with feature selection (5 features)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	ROC-AUC (%)
Hybrid 1D CNN-LSTM	**85.71**	**100.00**	57.14	72.73	76.53
Advanced 1D CNN	76.19	62.50	71.43	66.67	73.98
Advanced LSTM	71.43	55.56	71.43	62.50	67.35
SVM (RBF)	76.19	62.50	71.43	66.67	78.06
SVM (Linear)	80.95	66.67	**85.71**	**75.00**	**89.80**

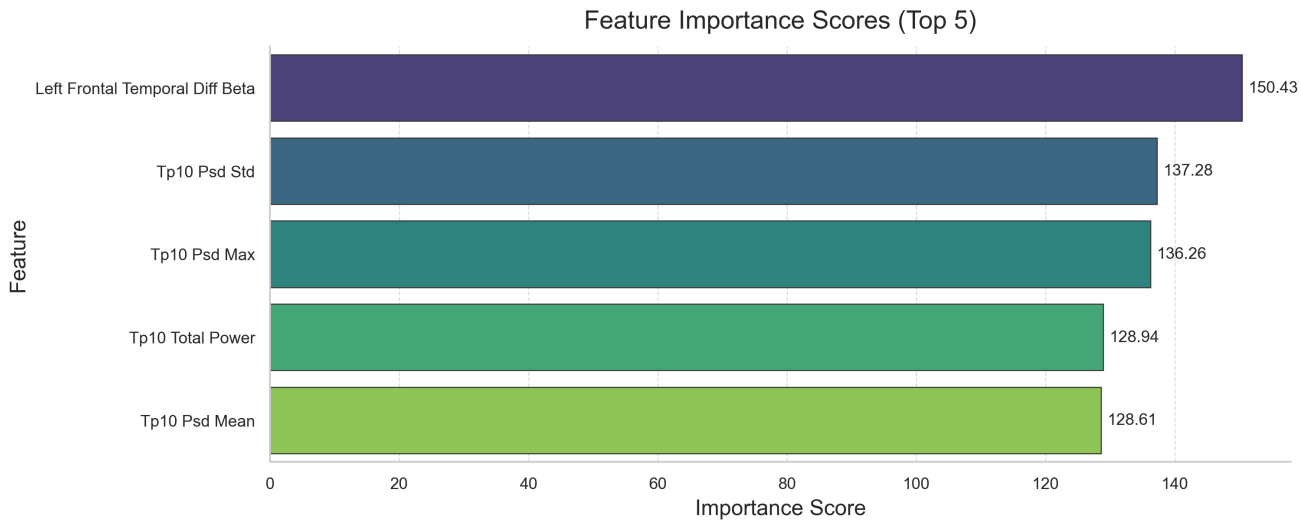


Figure 5: Top 5 Feature Importance Scores. The features are ranked by their importance score, highlighting the prominence of right temporal (TP10) features.

Table 4
Patient-level confusion matrices (counts) with feature selection

Model	TN	FP	FN	TP
Hybrid 1D CNN-LSTM	14	0	3	4
Advanced 1D CNN	11	3	2	5
Advanced LSTM	10	4	2	5
SVM (RBF)	11	3	2	5
SVM (Linear)	11	3	1	6

5-feature setting. See Table 7 and Figure 4c for metrics, and Table 8.

3.6. Hybrid Model Progression Across Feature Counts

The hybrid 1D CNN-LSTM shows the following performance progression as feature count increases (Table 9, Figure 6) and corresponding confusion matrices (Table 10, Figure 7).

3.6.1. Feature Selection Analysis

Across all models, moving from five selected features to the full 246-feature set resulted in large drops in accuracy, F1, and AUC (Tables 3 and 7). Notably, the Advanced LSTM and SVM (RBF) models failed to identify any responders without feature selection (TP=0), while the hybrid CNN-LSTM degraded substantially but retained minimal sensitivity (TP=1). These results indicate that feature selection is critical for this EEG classification task, likely mitigating the curse of dimensionality and focusing learning on the most informative temporal-spectral signatures.

4. Discussion

The hybrid 1D CNN-LSTM model achieved 85.71% accuracy for predicting tDCS treatment remission using a practical four-electrode EEG configuration, with perfect precision (100%), moderate recall (57.14%), F1-score 72.73%, and ROC-AUC 76.53%. This conservative prediction profile is clinically valuable: avoiding false positives reduces the risk of allocating patients to tDCS when they are unlikely to remit, while still correctly identifying more than half of true responders.

The dominance of TP10-related features in the selected feature set highlights the importance of right temporal region activity in tDCS response mechanisms. The selected features TP10 total power, TP10 PSD mean, TP10 PSD standard deviation, TP10 PSD maximum, and left frontal–temporal beta asymmetry align with prior evidence implicating temporal activity and inter-regional connectivity in depression pathophysiology and treatment response.

Our results significantly advance the limited body of research on EEG-based tDCS remission prediction. Al-Kaysi et al. [3] reported 76% mood prediction accuracy using traditional machine learning approaches in with 10 MDD patients [3]. Our 85.71% accuracy with 21 MDD participants represents both a performance improvement (9.71 percentage points) and a methodological advancement through hybrid deep learning architecture implementation.

Compared to recent meta-analyses of depression treatment prediction studies, our results demonstrate competitive performance with notable strengths. A meta-analysis of 15 studies with 479 patients reported an area under the curve of 0.91, sensitivity of 83% (95% CI 74–89%), and specificity of 86% (95% CI 81–90%) across various treatment modalities [9]. Our tDCS-specific results

Table 5
Patient-level performance with 20 selected features

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	ROC-AUC (%)
Hybrid 1D CNN-LSTM	61.90	33.33	14.29	20.00	59.69
Advanced 1D CNN	61.90	40.00	28.57	33.33	51.53
Advanced LSTM	52.38	20.00	14.29	16.67	27.55
SVM (RBF)	52.38	20.00	14.29	16.67	47.45
SVM (Linear)	47.62	16.67	14.29	15.38	36.73

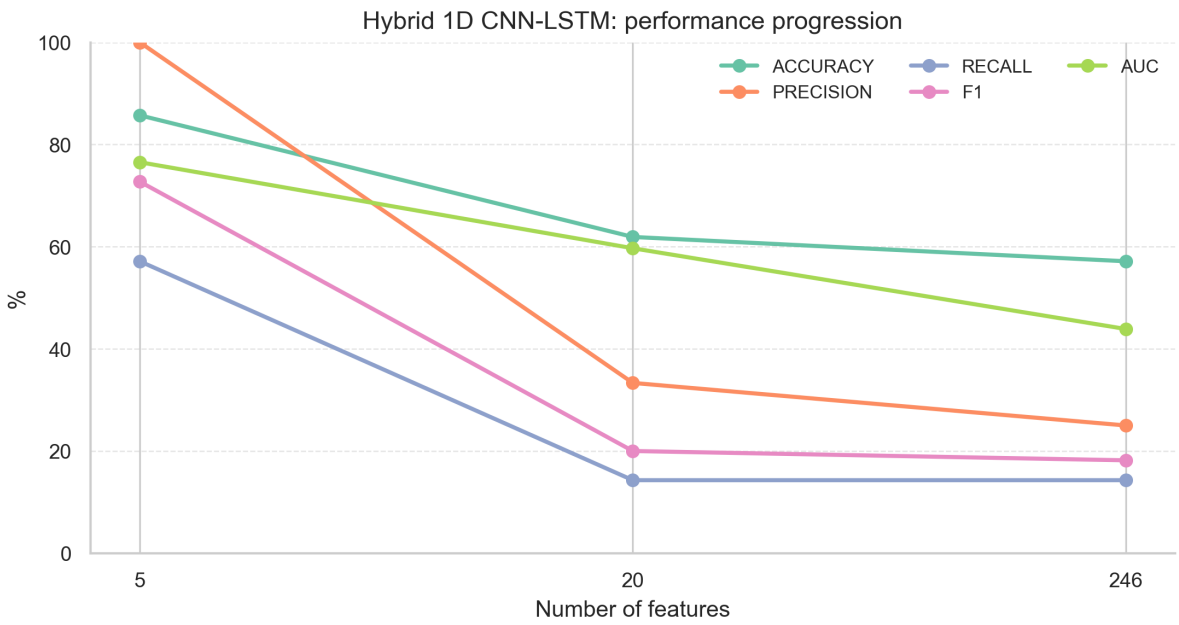


Figure 6: Hybrid 1D CNN-LSTM performance progression across feature counts (visual summary of Table 9).

(85.71% accuracy, 100% precision, 57.14% recall, F1-score 72.73, ROC-AUC 76.53) show strong precision performance, with high precision indicating that when our model predicts treatment remission, it is highly reliable. This conservative prediction profile, while achieving comparable overall accuracy to the broader literature, may be clinically advantageous for treatment selection applications where false positive predictions could lead to inappropriate treatment allocation.

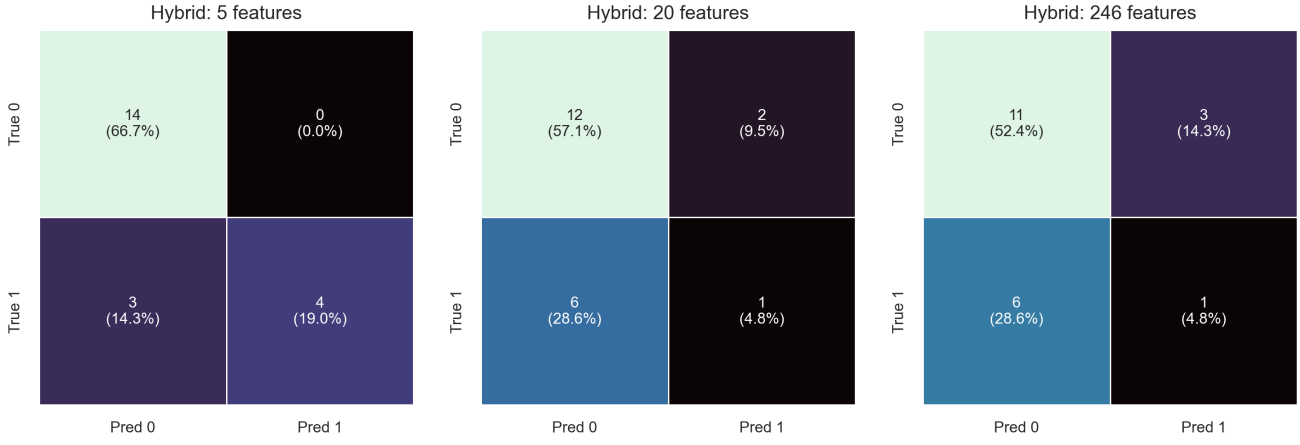


Figure 7: Hybrid 1D CNN-LSTM confusion matrices across feature counts (visual summary of Table 10).

Table 6
Patient-level confusion matrices (counts) with 20 selected features

Model	TN	FP	FN	TP
Hybrid 1D CNN-LSTM	12	2	6	1
Advanced 1D CNN	11	3	5	2
Advanced LSTM	10	4	6	1
SVM (RBF)	10	4	6	1
SVM (Linear)	9	5	6	1

In direct comparisons within our experiments using five selected features, SVM with a linear kernel achieved higher F1 (75.00%) and ROC-AUC (89.80%) than the hybrid model, whereas the hybrid model achieved the highest accuracy (85.71%) and perfect precision due to zero false positives. We selected the hybrid architecture as the primary approach because (i) its conservative prediction profile is preferable in clinical triage where false positives are costlier than false negatives, (ii) the architecture is inherently suited for temporal EEG, capturing sequential dependencies beyond static extracted features, and (iii) it maintained functionality across all feature counts, including extreme conditions (20 and 246 features), whereas some alternatives failed (e.g., TP=0 without feature selection).

The successful application of CNN-LSTM architecture to depression treatment prediction represents a novel contribution to the field. While hybrid architectures have achieved high accuracy in other EEG applications, their application to depression treatment response has been limited. Our results emphasize the critical role of feature selection: performance was optimal with 5 features (F1=72.73%), declined with 20 features (F1=20.00%), and further declined with all 246 features (F1=18.18%). These findings provide realistic, condition-wise benchmarks for clinical translation.

The broader EEG treatment prediction literature provides important context for our findings. Wu et al. [44] achieved R^2 values up to 0.60 for predicting continuous symptom improvement, demonstrating the potential for EEG-based regression approaches [44]. Zhdanov et al. [49] reported 79% accuracy using traditional machine learning with multiple EEG features, improving to 82% when incorporating early-treatment data [49]. Our approach's superior performance suggests that hybrid deep learning architectures may better capture the complex spatiotemporal patterns in EEG that characterize treatment response.

Our four-electrode configuration addresses a critical barrier to clinical translation that has limited previous research. Most EEG studies employ 19-128 electrode systems that are impractical for routine clinical use due to setup complexity, cost, and patient burden [7]. The successful prediction using only AF7, AF8, TP9, and TP10 electrodes represents a significant step toward deployable EEG-based treatment selection systems. This practical advancement could enable widespread adoption of personalized treatment selection in diverse healthcare settings.

Our comprehensive preprocessing pipeline addresses the temporal variability and artifact susceptibility that characterize clinical EEG data. Unlike controlled research settings, clinical EEG recordings contain substantial variability that must be carefully managed to achieve reliable results. Our approach demonstrates that robust preprocessing can enable effective learning from small, heterogeneous clinical datasets.

The integration of domain knowledge through hand-crafted features with deep learning-derived features represents a novel approach in this field. Most existing studies employ either purely traditional feature-based methods or end-to-end deep learning approaches. Our hybrid strategy leverages the complementary strengths of both: established neurophysiological markers provide interpretable, theory-driven features, while deep learning components capture complex patterns that may not be apparent to human experts.

The identification of specific predictive features provides valuable insights into the neurophysiological mechanisms underlying tDCS response. The prominence of TP10-related features suggests that right temporal region activity patterns serve as reliable biomarkers for treatment remission. This finding extends previous research on temporal lobe involvement in depression pathophysiology, which has identified altered temporal lobe metabolism and connectivity in depressed patients [39].

The temporal lobe's role in emotional processing and memory consolidation may be particularly relevant to tDCS mechanisms. tDCS effects on cortical excitability can influence network dynamics beyond the directly stimulated regions, potentially modulating temporal lobe activity through connected networks [43]. Patients with specific baseline temporal activity patterns may be more responsive to these network-level changes, explaining the predictive value of TP10 features.

The substantial contribution of left frontal-temporal beta band asymmetry (24.2% of model performance) underscores the importance of inter-regional connectivity patterns in treatment prediction. Beta band activity reflects cognitive control and attention regulation processes that are frequently impaired in depression and may be critical for tDCS treatment mechanisms [11]. The asymmetry component aligns with established theories of hemispheric imbalance in depression, where altered interhemispheric connectivity contributes to symptom expression [18].

These connectivity patterns may reflect the brain's capacity for neuroplasticity in response to stimulation. Patients with preserved or specific patterns of frontal-temporal connectivity may be better positioned to benefit from tDCS-induced plasticity changes. This interpretation suggests that baseline connectivity serves as a marker of neural reserve or plasticity potential, concepts that have been important in other neurological and psychiatric conditions [24].

Table 7
Patient-level performance without feature selection (246 features)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	ROC-AUC (%)
Hybrid 1D CNN-LSTM	57.14	25.00	14.29	18.18	43.88
Advanced 1D CNN	47.62	16.67	14.29	15.38	24.49
Advanced LSTM	47.62	0.00	0.00	0.00	33.16
SVM (RBF)	47.62	0.00	0.00	0.00	36.22
SVM (Linear)	52.38	20.00	14.29	16.67	34.69

Table 8
Patient-level confusion matrices (counts) without feature selection

Model	TN	FP	FN	TP
Hybrid 1D CNN-LSTM	11	3	6	1
Advanced 1D CNN	9	5	6	1
Advanced LSTM	10	4	7	0
SVM (RBF)	10	4	7	0
SVM (Linear)	10	4	6	1

The frequency-specific nature of our findings (particularly beta band contributions) aligns with research on tDCS's frequency-dependent effects. tDCS can modulate oscillatory activity across different frequency bands, with beta rhythms being particularly sensitive to stimulation parameters [31]. The predictive value of baseline beta activity may reflect individual differences in the responsiveness of these oscillatory networks to external modulation.

Analysing the same four-channel, home-EEG dataset as Moncy et al. [26], our study offers a complementary perspective and a distinct methodological contribution. Whereas their end-to-end deep models on PSD vectors emphasise a sensitivity-oriented operating point at comparable overall accuracy, we target a precision-oriented operating point by coupling a lightweight hybrid CNN–LSTM with attention to a compact, interpretable feature embedding spanning spectral statistics, inter-hemispheric asymmetry, and cross-channel relations. Under leave-one-patient-out evaluation, our model attains **85.71%** accuracy with **100% precision** at the patient level useful for confidently "ruling in" likely remission and minimising false positives while Moncy et al. [26] report **92.8%** sensitivity and **71.4%** specificity at **85.7%** accuracy, a profile well-suited to "ruling out" non-responders. Our novelty lies in explicit *feature parsimony* and interpretability: systematic ablations (5 vs. 20 vs. 246 features) show that a **five-feature** subset is sufficient and that larger sets can degrade calibration and F1, simplifying clinical translation. Notably, both analyses independently converge on *TP10* as informative Moncy et al. [26] via theta/alpha band power and ours via broader PSD statistics together with a left frontal–temporal beta asymmetry indicating aligned neuroanatomical evidence reached through different modelling choices [26].

Several limitations must be acknowledged when interpreting these results. The small sample size (n=21) limits statistical power and generalizability to broader populations. While our sample size aligns with current practice in the field (median 55 patients), larger validation studies are essential for confirming these findings and establishing clinical utility.

The four-electrode configuration, while practically advantageous, may miss important spatial information captured by high-density systems. Future research should investigate whether additional electrodes could improve performance without significantly compromising practical implementation. The current electrode selection was based on existing literature, but optimal electrode placement for tDCS prediction may differ from general depression biomarkers.

The study population was relatively homogeneous, limiting generalizability to diverse demographic groups and clinical presentations. Depression manifests differently across age groups, cultural backgrounds, and comorbidity profiles. Validation in more diverse populations is essential for establishing clinical utility across different patient groups.

The binary classification approach (remission vs. non-remission) simplifies the complex spectrum of treatment response. Future models could benefit from regression approaches that predict continuous improvement scores or multi-class classification that captures partial response categories.

Cross-validation within a single study site may not capture the full variability encountered in clinical practice. Multi-site validation would strengthen generalizability claims and address potential site-specific factors that could influence model performance.

Several research directions could build upon these findings to advance EEG-based treatment prediction toward clinical implementation. Larger multi-site studies are essential for confirming these results and establishing robust performance benchmarks. Collaborative efforts could pool data across research centers to achieve sample sizes necessary for definitive validation.

Investigation of optimal electrode configurations could balance performance with practical implementation. While four electrodes achieved good results, systematic evaluation of electrode additions could identify the minimal configuration necessary for maximal performance.

Table 9
Hybrid 1D CNN-LSTM performance across feature counts

Features	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	ROC-AUC (%)
5 (selected)	85.71	100.00	57.14	72.73	76.53
20 (selected)	61.90	33.33	14.29	20.00	59.69
246 (no FS)	57.14	25.00	14.29	18.18	43.88

Table 10
Hybrid 1D CNN-LSTM confusion matrices across feature counts

Features	TN	FP	FN	TP
5 (selected)	14	0	3	4
20 (selected)	12	2	6	1
246 (no FS)	11	3	6	1

Advanced electrode selection algorithms could optimize placement for individual patients or clinical populations.

Looking ahead, multimodal predictors could be explored for example, (i) using HYDRA-derived MRI dimension membership or hyperplane distances as complementary markers [13], (ii) probing whether EEG-based predictors differ across dimensions, and (iii) comparing practical fusion schemes (early, late, shared-representation) to assess whether combining EEG with MRI might improve calibration and generalisability while preserving a deployable footprint.

Integration of multimodal biomarkers could enhance prediction accuracy beyond EEG alone. Combining EEG with neuroimaging, genetic markers, clinical assessments, and behavioral measures might capture complementary aspects of treatment response mechanisms. Machine learning approaches could identify optimal feature combinations across modalities.

Longitudinal studies tracking patients throughout treatment could provide insights into response trajectories and enable early prediction of treatment outcomes. Current approaches predict final response status, but early identification of non-response could enable treatment modifications before completing ineffective protocols.

Development of interpretable AI approaches could enhance clinical adoption by providing mechanistic insights into predictions. While our feature importance analysis provides some interpretability, advanced explainable AI methods could offer more detailed insights into decision-making processes that clinicians could understand and trust.

Real-world validation studies in clinical settings are essential for translating research findings into practice. Laboratory-based studies may not capture the full complexity of clinical implementation, including workflow integration, user acceptance, and real-world performance variability.

5. Conclusion

This study presents the first application of hybrid 1D CNN-LSTM architecture for predicting tDCS treatment remission in major depressive disorder, achieving 85.71% accuracy using a practical four-electrode EEG configuration. The results demonstrate competitive performance with existing depression treatment prediction models while addressing critical gaps in tDCS-specific research and clinical implementation feasibility.

The key contributions include establishing the feasibility of accurate tDCS remission prediction using deep learning approaches, demonstrating effective prediction with minimal electrode requirements suitable for clinical deployment, and identifying specific EEG features predictive of treatment remission. The model's conservative prediction profile, characterized by perfect precision and moderate recall, aligns well with clinical requirements for treatment selection applications.

The identification of right temporal region activity patterns as primary predictors extends our understanding of tDCS response mechanisms and provides clinically relevant biomarkers for treatment optimization. The importance of frontal-temporal connectivity patterns further highlights the role of baseline network characteristics in determining treatment outcomes.

While limitations including small sample size and single-site validation must be acknowledged, this work establishes a foundation for larger validation studies and clinical translation efforts. The successful demonstration of accurate prediction using practical electrode configurations represents a significant step toward implementing precision psychiatry approaches in routine clinical care.

Future research should focus on multi-site validation, optimization of electrode configurations, integration of multimodal biomarkers, and real-world clinical implementation studies. The ultimate goal of developing clinically deployable treatment selection tools requires continued advancement of both predictive accuracy and practical implementation feasibility.

This research contributes to the growing body of evidence supporting EEG-based biomarkers for depression treatment selection while specifically addressing the underexplored area of tDCS remission prediction. The combination of methodological rigor, practical clinical considerations, and competitive performance positions this work as a valuable contribution toward achieving precision psychiatry implementation for improving depression treatment outcomes.

`\printcredits` command is used after appendix sections to list author credit taxonomy contribution roles tagged using `\credit` in frontmatter.

CRediT authorship contribution statement

Abin Daniel Zorto: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.. **Jijomon C. Moncy:** Writing - review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization.. **Mhd Saeed Sharif:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Cynthia H.Y. Fu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

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